INCH - POUND

MIL-T-47248A(MI)
13 February 1991
SUPERSEDING
MIL-T-47248(MI)
26 July 1974

MILITARY SPECIFICATION
TAPE, PLASTIC, HEAT-SEALING, PERFORATED

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes a fiberglass - reinforced plastic, perforated, heat-sealing, tear-away tape.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135

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SPECIFICATIONS

FEDERAL

UU-T-450	-	Tissue, Facial
MMM-A-121	-	Adhesive, Bonding Vulcanized Synthetic Rubber to Steel

MILITARY

MIL-P-5425	-	Plastic, Sheet, Acrylic, Heat Resistant
MIL-I-47100	-	Insulation Sleeving Flexible, Braided, 200 Degrees Celsius (High Temperature Polyamide)

STANDARDS

MILITARY

MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MIL-STD-2073-1	-	DoD Materiel Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	-	Packaging Requirement Codes

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740	-	Standard Specification for Methyl Ethyl Ketone
ASTM D 1000	-	Pressure - Sensitive Adhesive Coated Tapes Used for Electrical Insulation, Methods of Testing

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. *REQUIREMENTS*

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material.

3.2.1 Composition and physical properties.

3.2.1.1 Composition. The product shall be a fiberglass-reinforced plastic tape coated with polyamide (nylon) compound, with perforations down the centerline. It shall be capable of heat sealing to itself by the application of heat and pressure.

3.2.1.2 Color. The tape shall be black in color and opaque.

3.2.1.3 Form. Unless otherwise specified (see 6.2), the tape shall be furnished in rolls having a 2-inch core diameter. Splices with overlaps which do not exceed 0.070 inch wide and which provide tear strength in accordance with 3.3.2 are allowed.

3.2.2 Dimensions. The tape width shall be 1.125 $\pm$ 0.005 inches, and the thickness shall be 0.004 $\pm$ 0.0003 inches. The thickness requirement does not apply to spliced areas.

3.2.3 Perforations. The tape shall have perforations along its centerline within the dimensional tolerances shown in figure 1. All material shall be removed from the perforations, except that individual cut fibers may cling to edge.

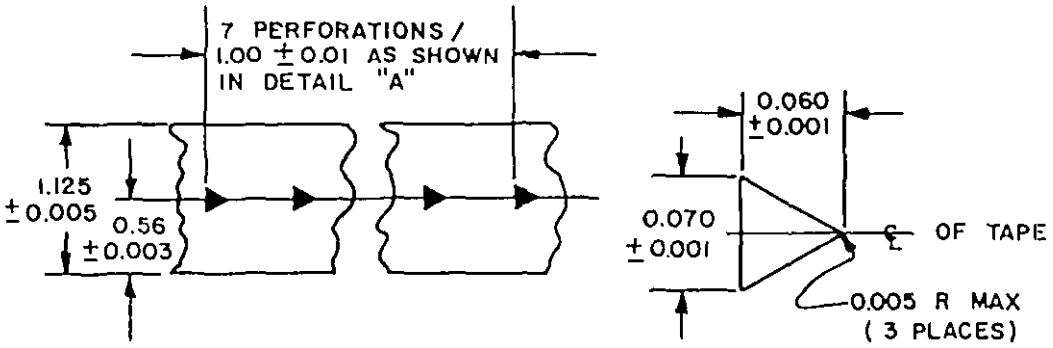
3.2.4 Storage life. The tape as furnished by the manufacturer shall be capable of meeting the requirements of this specification after storage in its original package at 16° to 32° Celsius (C).

3.3 Performance characteristics.

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3.3.1 Heating sealing. The tape shall be capable of forming a strong, uniform bond to itself, and the surface above the bond shall appear glossy.

3.3.2 Tear-away strength. The tape shall have an average tear-away strength along the line of perforations of 1.0 to 3.0 pounds at temperatures of -32° ± 3°C, 25° ± 3°C, and 60° ± 3°C.



DETAIL A PERFORATION

Note: All dimensions are in inches.

FIGURE 1. Tape perforations.

3.4 Environmental conditions.

3.4.1 Chemical resistance. The tape shall show no evidence of damage or softening when tested with methyl ethyl ketone (MEK) conforming to ASTM D 740 as specified herein.

3.4.2 Fungus resistance. The tape shall support no fungus growth visible to the unaided eye, when tested as specified herein.

3.4.3 Heat resistance. The tape shall exhibit no visible distortion or change in dimensions when heated as specified herein.

3.5 Workmanship. The workmanship shall be such as to ensure the tape is free from flaws, defects, pinholes, bubbles, cracks, or inclusions.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When specified (see 6.2), a sample consisting of 1 roll of 25 feet of tape, manufactured under the same conditions as those proposed for subsequent production, shall be subjected to first article inspection. Accomplishment of first article inspection shall be as specified herein. If the first article sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained. Units subjected to first article inspection shall have successfully passed acceptance inspection. Testing of the first article sample, to determine compliance with the characteristics listed in table I, shall be conducted in accordance with the corresponding test paragraphs.

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TABLE I. First article inspection.

Characteristic	Requirement Paragraph	Test Paragraph
Composition and physical properties	3.2.1	4.6.3.1
Dimensions	3.2.2	4.6.3.2
Perforations	3.2.3	4.6.3.3
Storage Life	3.2.4	4.6.3.4
Heat sealing	3.3.1	4.6.3.5
Tear-away strength	3.3.2	4.6.3.6
Chemical resistance	3.4.1	4.6.3.7
Fungal resistance	3.4.2	4.6.3.8
Heat resistance	3.4.3	4.6.3.9
Workmanship	3.5	4.6.4

4.4 Quality conformance inspection.

4.4.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be in accordance with table II. A roll of tape shall be considered a unit of product. Randomization of the containers in the lot, as well as randomization of the location and number of rolls (or units) within each container, shall be employed when sampling for quality conformance.

TABLE II. Sampling plan.1/

Lot size	Sample size
2 to 8	100%
9 to 90	8
91 to 150	12
151 to 280	19
281 to 500	21
501 to 1200	27
1201 to 3200	35
3201 to 10,000	38
10,001 to 35,000	46
35,001 to 150,000	56
150,001 and over	64

1/Acceptance number is "ZERO."
Reject number is "ONE."

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4.4.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.4.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.4.1, to determine compliance with the characteristics specified in table III, shall be conducted in accordance with the corresponding method and inspection paragraphs.

TABLE III. Quality conformance inspection.

Characteristic	Requirement Paragraph	Method Paragraph
Composition and physical properties	3.2.1	4.6.3.1
Dimensions	3.2.2	4.6.3.2
Perforations	3.2.3	4.6.3.3
Heat sealing	3.3.1	4.6.3.5
Tear-away strength at 25°C	3.3.2	4.6.3.6
Workmanship	3.5	4.6.4
Packaging, packing, and marking	5	5.1, 5.2

4.5 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient ± 9°C

b. Altitude, facility ground

c. Humidity, facility ambient up to 95 percent relative humidity

4.6.2 Test sequence. Unless otherwise specified (see 6.2), testing sequence within each classification of inspection (first article or quality conformance) shall be at the option of the supplier.

4.6.3 Tests.

4.6.3.1 Composition and physical properties. Composition shall be certified by the supplier and the physical properties shall be examined visually to establish compliance to 3.2.1.

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4.6.3.2 Dimensions. The width shall be determined by the use of a suitable instrument graduated to read within 0.001 inch. The tape shall be flat on a smooth surface at the time of measurement. At least 10 measurements shall be made. The thickness shall be determined in accordance with ASTM D 1000. Dimensions shall conform to 3.2.2.

4.6.3.3 Perforations. A specimen of tape shall be placed between 0.17 inch sheets of heat resistant acrylic plastic conforming to MIL-P-5425. The dimensions and spacing of perforations shall then be measured with a Kodak Optical Comparator, or equivalent, and shall be as specified in 3.2.3.

4.6.3.4 Storage life. One roll of tape in its original package shall be stored under the conditions specified in 3.2.4. At the end of the storage period, the tape shall be subjected to all the tests contained herein. The supplier's certificate of compliance certifying to the storage life requirements (see 3.2.4) may be accepted during the storage period.

4.6.3.5 Heat sealing. A length of tape at least 6 inches long shall be folded double along its centerline. The doubled area shall be pressed with a heated tool maintained at a surface temperature between 182° and 204° C. The time of contact of the tool with the tape shall not exceed 1 second. The temperature shall be adjusted in such a manner as to cause the overlapped layers of tape to bond together without blistering or burning of the tape. A correct bond shall have strength sufficient to prevent pulling the layers apart without tearing the tape and shall conform to 3.3.1.

4.6.3.6 Tear-away strength. A polyamide (nylon) sleeving conforming to MIL-I-47100, of approximately 0.035 inch outside diameter with 6 0.005 inch steel wires enclosed, shall be placed along the perforated line on the tape. The tape shall be folded over at the perforation line and heat sealed as specified in 4.6.3.5, with the sleeving encapsulated at the perforation line. The full length of the tape shall then be bonded with adhesive conforming to MMM-A-121 or clamped by a suitable means to a flat aluminum plate. A length of sleeving which has been left extending out of the tape retainer shall be used to apply a pulling force parallel to the perforation line approximately in a plane with the tape as bonded or clamped to the plate. The angle created between the sleeving being pulled and the perforation line shall be approximately 10 to 15 degrees. The average tear-away force (i.e., force causing the sleeving to tear out of the perforations) shall be measured as the sleeving is being torn away at a rate of 5 to 15 inches per second, at each of 3 temperatures: -32° $\pm$ 3°C; 25° $\pm$ 3°C; and 60° $\pm$ 3°C, and shall be as specified in 3.3.2.

4.6.3.7 Chemical resistance. Chemical resistance of the tape shall be tested as specified below and shall conform to 3.4.1.

Place a small pad of tissue conforming to UU-T-450, approximately 1 inch square and 8 to 12 layers thick, on the tape surface. Fully saturate with MEK

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conforming to ASTM D 740. Cover the pad with a watchglass or small beaker. Remove the pad after 1 minute and rub the moistened spot with a clean dry cloth or tissue. Failure is indicated by discoloration of the cloth or tissue or by visible evidence of damage to the tape surface. A slight whitening of the surface which may be removed by light rubbing with a dry cloth is not reason for rejection. Resistance to softening is determined by scratching the surface with a fingernail.

4.6.3.8 Fungus resistance. The tape shall be tested for fungus and mildew resistance in accordance with MIL-STD-810, Method 508.4, and shall be as specified in 3.4.2.

4.6.3.9 Heat resistance. A short length of tape shall be held horizontally above the flame of a match or alcohol burner in such a manner that the plastic matrix is ignited and burns. After burning ceases, the tape shall be visually inspected for evidence of warping, curling, or change of size. The results shall be as specified in 3.4.3.

4.6.4 Workmanship. The tape shall be visually examined to assure conformance to the workmanship requirement specified in 3.5.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-STD-2073-1 and MIL-STD-2073-2.

5. PACKAGING

5.1 Preservation, packaging and packing. Packaging requirements shall be in accordance with the following MIL-STD-2073-1 and MIL-STD-2073-2 codes:

		MP	CD	PM	WP	CD	CT	UC	IC	PK	SM
5.1.1	Level A.	2D	1	00	00	00	0	XX	XX	E	99
5.1.2	Level B.	2D	1	00	00	00	0	XX	XX	A	99
5.1.3	Level C.	10	1	00	YY	00	0	YY	YY	L	99

5.2 Marking. Marking shall include, but be not limited to, the following information:

a. Batch or lot number

b. Date of manufacture (month and year)

c. Shelf life expiration date

d. Number and date of this specification

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This tape is intended for use as a tear-away fastening device for positioning lead wires on the exterior of missiles. It is suitable for use between -32° and 60°C after storage between -62° and 68°C.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Whether a first article is required (see 3.1 and 4.3)

d. Form, if different from 3.2.1.3

e. Requirement for certification of composition (see 4.6.3.1)

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

a. Adhesive, strip

b. Cement

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c. Glue

d. Mucilage

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing Activity:
Army-MI

Project No. 8135-A627

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-47248A (MI)	13 February 1991

3. DOCUMENT TITLE
TAPE, PLASTIC, HEAT-SEALING, PERFORATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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